

Thesis progress report

LLM-Guided Neural Network Compilation for Real-time UAV Edge Inference

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A hosted model, one revision

The plan already has to pass a gate, and only a measured plan is kept. The planner itself can be a model on a remote host.

There is one HTTP client. Groq is one named host. OpenAI, Anthropic, Gemini, DeepSeek, and other hosts that speak the same chat API are further rows. A new host is a URL and a model id. The API key stays outside git.

After a plan has been measured, there is at most one extra call. If compile or the numeric check failed, that call is named `llm_revised`. If the plan passed but was strictly slower than the best legal plan, the call is named `llm_improved`. A rate limit is stored as the text LLM HTTP 429. The extra plan is allowed to lose.

A slow reasoning model needs a longer timeout and sometimes no temperature field. Those are settings in the environment, not a new program. The same three tries from the gate still apply on that extra call. Only one extra compile is run.

Groq was hard-coded at first. A second host would have meant a second program, and a rate limit would have looked like a missing result. Pose hit HTTP 429 on the first call. The row is the heuristic, and the reason is stored. MobileNet hit 429 on the extra try. That compile was skipped instead of saving the heuristic under the name of a revision. The extra try itself used to die on one bad reply. It now gets the same three tries as the first plan, and it still may compile only once. The model is allowed to lose. That is the point of writing the rank down.

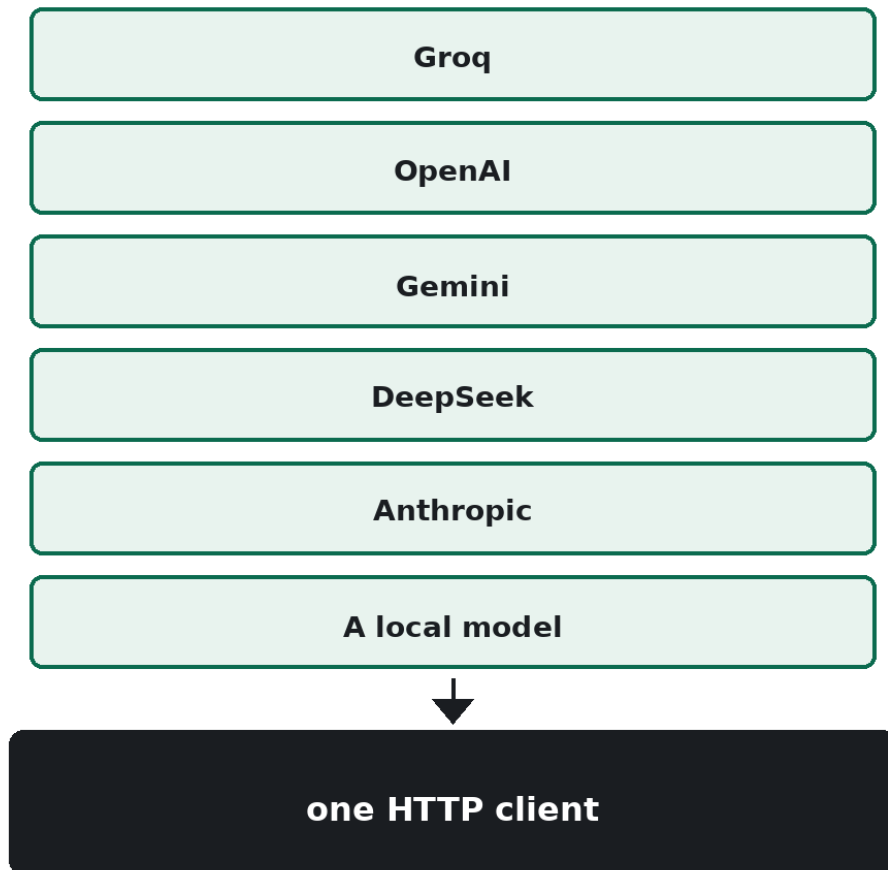


Figure 1. Many hosts, one client, then one extra measured try.

A remote model can name a plan, and the record can show that the plan lost. Running that loop by hand is still a long sequence of commands.